

Math: Asymptotic (Big-O/ Θ / Ω) Notation in a Single Variable

Note 1. We will use asymptotic notation to measure properties of our programs like: runtime, memory usage, and API cost. In future classes, you will also be using asymptotic notation for measuring statistical properties of programs as well like how accurate a machine learning algorithm is based on the number of data points trained on. We will therefore use a slightly technical definition of asymptotic notation that can be used in all of these scenarios. But we will focus on intuition and not get into weird mathematical technicalities that don't come up in practice.

Definition 2. Let f, g be functions from $\mathbb{R}^+ \rightarrow \mathbb{R}^+$. Then,

1. If $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} < \infty$, then we say $f = O(g)$.
2. If $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} > 0$, then we say $f = \Omega(g)$.
3. We say that $f = \Theta(g)$ if both $f = O(g)$ and $f = \Omega(g)$.

Intuitively, you should think of O as $\leq$, Ω as $\geq$, and Θ as $=$.

Example 3.

1. $f(x) = x$
 $g(x) = x^2$

2. $f(x) = x^2$
 $g(x) = x$

3. $f(x) = x^2 + 2x + 5$
 $g(x) = x$

4. $f(x) = x^2 + 2x + 5$
 $g(x) = x^2$

5. $f(x) = x^2 + 2x + 5$
 $g(x) = x^3$

Note 4. The main idea of asymptotic notation is that we can “discard” the “small” terms and only focus on the “large” terms. This makes working with functions much simpler. We lose “accuracy” for small numbers, but in practice the behavior of large numbers is much more important.



Example 5. It is often useful to simplify asymptotic expressions. In this example, we will work out how to simplify

$$O\left((2n+1)^3\right)$$

The procedure is first expand the expression into a sum of terms:

$$(2n+1)^3 = 8n^3 + 12n^2 + 6n + 1$$

Replace leading constants with 1:

$$\sim n^3 + n^2 + n + 1$$

Discard every term that is O of another term:

$$\sim n^3$$

So our initial expression simplifies to $O(n^3)$. This procedure works for O , Θ , and Ω .

Example 6. Simplify the following expressions:

1. $O\left(8n^3 + n^2\right)$

2. $\Omega\left((7n)^3 + 5n^2 \log n + \log n\right)$

3. $\Theta\left(1000000000000\right)$

4. $O\left(\log n + 1000000000000\right)$

5. $O\left(\frac{1}{n} + \frac{1}{n^2}\right)$